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Visual Intelligence

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CLFormer: a cross-lingual transformer framework for temporal forgery localization

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Abstract

Temporal forgery localization (TFL) is crucial in deepfake detection. It focuses on identifying subtle temporal manipulations within video content. However, the generalization capabilities of current TFL methods are limited, especially across different languages, which limits their performance in diverse environments. This limitation stems from two key factors. First, most existing datasets are English-centric. Second, there is inadequate learning from multi-modal information, where visual features are often prioritized over audio features. To address this gap, we created the Chinese audio-visual deepfake (CHAV-DF) dataset, which is the first dataset designed for the TFL in the Chinese context. This dataset provides a valuable benchmark for evaluating the TFL methods in cross-lingual settings. Additionally, we introduced a cross-lingual transformer framework (CLFormer), which prioritizes audio features and utilizes a pre-trained multi-lingual Wav2Vec2 to enhance cross-lingual generalization, while incorporating visual features to further refine TFL. Moreover, we incorporated a refinement module into CLFormer to enhance the accuracy of forgery localization. Experiments on the LAV-DF, CHAV-DF, and AV-Deepfake1M datasets demonstrate that CLFormer performs well in both same-language and cross-language settings. Specifically, CLFormer achieves an average precision (AP) of 57.68% at temporal intersection over union (tIoU) of 0.50 when trained on CHAV-DF and tested on LAV-DF, surpassing the state-of-the-art method by 47.59%, and validating its cross-language generalization capability.

Keywords: Temporal forgery localization (TFL), Cross-lingual, Audio feature, Wav2Vec2, Boundary refinement

1 Introduction

The advancement of generative technology [1, 2] has made deepfake technology increasingly prevalent, resulting in significant security and ethical challenges due to its high realism and diverse applications. Detecting multi-modal content forgeries is especially difficult because of their complexity, underscoring the urgent need for more effective detection methods.

Current video deepfake detection methods rely primarily on binary classification, which struggles with partial manipulations. Temporal forgery localization (TFL) addresses this issue by detecting tampered segments within videos, enhancing detection granularity, and providing insights into the timing of manipulations. However, the effectiveness of existing methods [3–6] is significantly constrained by their generalization capabilities, particularly in cross-lingual contexts. This limitation arises from two key factors. First, there are currently only two datasets [3, 7] specifically designed for the TFL task, both of which are English-centric. This lack of dataset diversity restricts the generalization of different languages, leading to suboptimal performance in various environments. Second, many existing methods extend from temporal action localization tasks that typically emphasize visual features and then merely concatenate simple audio features. This approach tends to ignore the effect of linguistic diversity on audio features, resulting in weak generalization for locating forgery videos in different languages. To address these

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challenges, we construct a Chinese audio-visual deepfake (CHAV-DF) dataset, which is the first dataset designed specifically for the TFL task in the context of the Chinese language. CHAV-DF includes a range of audio and video forgeries, setting a new benchmark for deepfake detection. By constructing CHAV-DF, we can explore the impact of different audio features and evaluate the generalization capabilities of existing techniques across datasets and languages.

To enhance localization performance across different languages and improve generalization, we propose a novel cross-lingual transformer framework (CLFormer). To address language differences, such as those between Chinese and English, we incorporate a pre-trained multi-lingual Wav2Vec2 [8] to extract audio features from various datasets. This approach helps capture the commonality of audio features across languages and improves adaptation to multi-lingual contexts. Additionally, we introduce a multi-level refinement module to compute offsets at each time point and refine the TFL results through post-processing. This method allows CLFormer to achieve high accuracy on individual datasets and demonstrates enhanced generalization across datasets. The main contributions of this paper are as follows:

1) A Chinese audio-visual deepfake dataset called CHAV-DF is constructed to provide a benchmark in the Chinese language, enhancing the evaluation of multi-modal TFL across diverse linguistic contexts.
2) We propose a cross-lingual transformer framework termed as CLFormer that integrates a multi-lingual Wav2Vec2 with a refinement module. Our approach effectively leverages audio features and refines results, enhancing cross-lingual TFL performance.
3) Experimental results show that CLFormer excels in the same-language and cross-lingual settings, achieving average precision (AP) of 57.68% at temporal intersection over union (tIoU) of 0.50 on LAV-DF [3] when trained on CHAV-DF, surpassing the state-of-the-art by 47.59%.

2 Related work

2.1 Deepfake detection datasets

The rapid advancement of generative technologies has propelled the field of deepfake detection, particularly in binary classification tasks. Traditional datasets have focused on scenarios in which the entire video is fabricated, such as those created through face swapping. These datasets have been foundational in advancing deepfake detection capabilities.

To address the growing complexity of forgery techniques and the need for detailed detection, LAV-DF is introduced as the first content-driven audio-visual dataset designed specifically for TFL. LAV-DF represents a major advancement by addressing the limitations of traditional datasets and focusing on temporal aspects of forgery localization. Subsequently, AV-Deepfake1M [7] is developed to bridge further gaps in dataset quality, diversity, and size. AV-Deepfake1M utilizes advanced text, video, and audio generation techniques to construct a large-scale audio-visual dataset, setting a new benchmark for TFL. Despite these advances, both LAV-DF and AV-Deepfake1M are primarily English-language datasets, which limits their effectiveness in multi-lingual contexts. This constraint affects the detection of forged content in non-English languages and limits its applicability in diverse real-world scenarios.

2.2 Temporal forgery localization

TFL aims to identify the start and end timestamps of forgeries in a video, which requires both forgery detection and temporal localization. Early approaches such as BA-TFD [3] utilize convolutional neural networks (CNNs) for audio and visual feature extractions, combined with boundary-aware detection mechanisms for localization. BA-TFD+ [4] improves upon this by incorporating multi-scale vision transformers, which enhance boundary detection. Recently, AV-TFD [9] introduces a model that consists of encoders with cross-modal attention and an embedding level fusion mechanism. MABC-Net [5] integrates mixed-attention feature learning and fusion-balanced localization modules to effectively address the class imbalance issue and enhance the accuracy of TFL. However, these methods still face challenges, such as efficiently encoding audio-visual features, which leads to increasing processing times and less effective localization accuracy.

In response to these challenges, UMMAFormer [6] incorporates a transformer architecture with a temporal feature abnormal attention module and a parallel cross-attention feature pyramid network. This approach enhances the representation of audio-visual features and enables more precise detection of forged fragments. However, despite these advancements, handling audio features in UMMAFormer [6] remains basic. DiMoDif [10] leverages lip reading and speech-to-text models to extract visual and audio features, and a transformer encoder with local attention to detect frame-level cross-modal incongruities for TFL. MMMS-BA [11] adopts a multi-modal attention framework based on recurrent neural networks (RNNs), utilizing contextual information from audio-visual sequences for deepfake detection and localization.

Recent advances in the TFL task have been demonstrated in the 1M-deepfakes detection challenge [12], where several methods [13–16] have emerged. Vigo [13] leverages a multi-layer perceptron approach to process both audio and video features for deepfake localization. MFMS [14] introduces novel modality-fusion and modality-specific modules built upon the UMMAFormer [6]

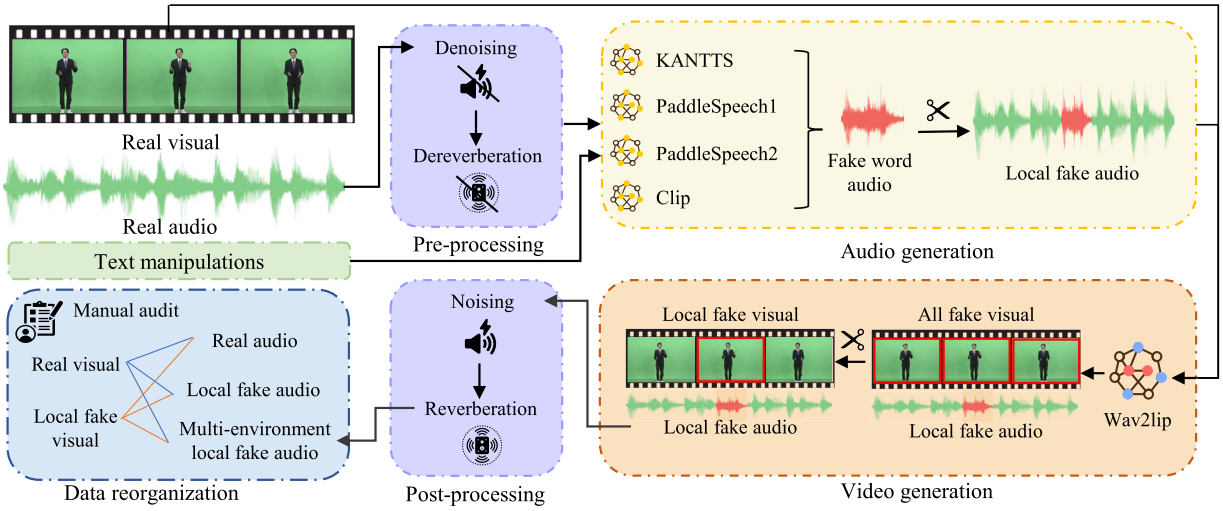

Figure 1 Generation pipeline of the Chinese audio-visual deepfake (CHAV-DF) dataset. Text generation is performed by modifying adjectives, quantifiers, verbs, and nouns. Audios are generated via text-to-speech (TTS) and clipping techniques to synthesize fake speech. The TTS technologies include PaddleSpeech1 [17], PaddleSpeech2 [17], and KANTTS [18]. The synthesized audio is then used for video generation via Wav2Lip [19] to ensure visual-lip synchronization. Finally, all the components are assembled into the dataset, followed by a quality screening process. The images are sourced from the CUC-RED [20] dataset

framework to enhance its localization capabilities. THE-FD [15] presents a task hierarchical emotion-aware architecture with an adaptive audio-video fake-frame heatmap module to improve temporal localization. W-TDL [16] employs a window-based audio approach combined with a complementary visual frame-based model to increase the accuracy of TFL. Overall, existing methods exhibit notable shortcomings in integrating and processing audio-visual features, particularly in the inadequate utilization of audio features. This limitation hampers their cross-lingual generalization ability, affecting the accuracy and robustness of forgery localization.

3 Chinese audio-visual deepfake dataset

This section details the construction of CHAV-DF, including data collection, processing, and dataset statistics, as shown in Fig. 1.

3.1 Data collection

Our real video data are sourced from the CUC-RED [20] dataset. The dataset contains real video clips recorded in a green screen studio, featuring 12 broadcasting and hosting students (5 female, 7 male). Each student reads 90 sentences designed to convey happy, sad, and neutral emotions. The sentences range from 10 to 20 words in length, resulting in a total of 1080 real video samples.

3.2 Data processing

Text generation For the 90 original text statements, tampering is performed in two ways. First, we modify adjectives, quantifiers, verbs, and nouns with similar or opposite meanings. Second, words from the original statement are replaced with those from other statements. Throughout this process, maintaining sentence fluency is our priority.

Audio generation After modifying the text, we utilize three text-to-speech (TTS) technologies and clipping techniques to generate fake word audio. Many advanced TTS models struggle with generating high-quality Chinese audio, which affects the fidelity of local fake audio. To ensure optimal performance in the Chinese context, TTS models pre-trained on Chinese datasets are employed to synthesize voices for specific identities. The models used include PaddleSpeech1 [17] (FastSpeech2 [21]-HiFiGAN [22]), PaddleSpeech2 [17] (FastSpeech2 [21]-PWGAN [23]), and KANTTS [18] (Sambert-HiFiGAN [22]). First, denoising and dereverberation are applied to the original real audio. Then, fake word audio is generated via TTS technology, replacing the corresponding segments. For the clipping process, matching audio segments of the same identities are identified based on the modified text, clipped, and stitched together to produce local fake audio.

Video generation We proceed to generate the corresponding fake visuals when the fake audio is prepared. Since full-body shots and background environments are included in our videos, Wav2Lip [19] is used to preserve the additional context in the video. The full audio drives the video generation, and we clip video frames corresponding to the fake word audio. These frames are then

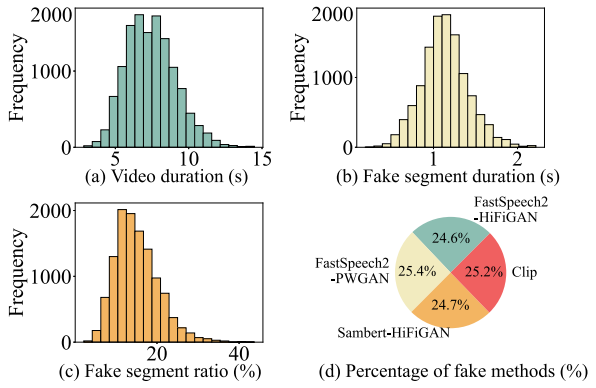

Figure 2 Distribution of the CHAV-DF dataset. (a) Video duration: video lengths in seconds. (b) Fake segment duration: manipulated segment lengths in seconds. (c) Fake segment ratio: percentage of manipulated segments relative to the video length. (d) Percentage of fake methods: proportions of audio manipulation methods, including three methods utilizing TTS technology: PaddleSpeech1 [17] (FastSpeech2 [21]-HiFiGAN [22]), PaddleSpeech2 [17] (FastSpeech2 [21]-PWGAN [23]), and KANTTS [18] (Sambert-HiFiGAN [22]), as well as a method based on audio clip editing

replaced in the original video to create local fake visuals. To simulate diverse environments, two noisy and reverberant versions of local fake audio are created as additional references and are not included in the main dataset. After data generation, we manually screen out videos with poor visual or audio quality and then combine all the data to assemble the final dataset.

3.3 Dataset statistics

CHAV-DF consists of 12,951 videos with diverse content and characteristics. Among these, 1080 videos are authentic, whereas 11,871 videos contain one manipulated segment. The dataset, totaling 27.78 h, is divided into three subsets: 9099 videos for training, 1920 videos for validation, and 1932 videos for testing. On average, each video is 7.72 s long, with manipulated segments averaging 1.19 s. Figure 2 provides insights into the distribution of video durations and manipulated segments, illustrating the characteristics of the dataset.

4 The proposed method

4.1 Overview

To enhance cross-lingual generalization, we propose the CLFormer, as illustrated in Fig. 3. The framework operates in three stages: 1) extracting visual features $F_v \in \mathbb{R}^{C_v \times T}$ with a TSN [24] backbone and audio features $F_a \in \mathbb{R}^{C_a \times T}$ using Wav2Vec2 [8] from the video sequence X, and concatenating these features. C_v and C_a denote the dimensions of the visual and audio feature channels, respectively, and T is the number of temporal segments obtained by evenly splitting the input sequence X; 2) performing initial forgery detection and localization with a TFL detector; and 3) refining the detection results with a refinement module to produce the final output $S' = \{(t_n^{s'}, t_n^{e'}, s'_n)\}_{n=1}^{N}$, where $t_n^{s'}$ and $t_n^{e'}$ are the refined start and end times of the n-th forged segment, s'_n is the corresponding confidence score, and N is the total number of predicted forged segments.

4.2 Pre-trained feature extractor

In the TFL task, the importance of audio features is often overshadowed by a pre-dominant focusing on visual information. This limitation becomes particularly evident in cross-lingual scenarios where linguistic differences impact generalization. To overcome this, CLFormer utilizes Wav2Vec2 [8] for audio feature extraction. Pre-trained on over 500,000 h of real-world speech, this model distinguishes authentic from forged audio by extracting features with greater distinctiveness. These unique properties enable the detection of subtle discrepancies often missed by visual analysis alone, significantly enhancing the cross-lingual adaptability and overall effectiveness of the TFL methods. Our approach to audio feature extraction in CLFormer involves two key components: feature encoder and context network.

Feature encoder The feature encoder processes raw audio A, producing a sequence of latent speech representations Z. It comprises seven 1D convolutional layers, each followed by layer normalization and a gaussian error linear unit activation function. These representations are extracted from real audio during training and naturally show significant differences when the input is manipulated, thus providing important clues for identifying forged segments.

Context network The latent representations Z generated by the feature encoder are further processed by a context network based on the transformer architecture. This network consists of 24 layers, each with 16 attention heads and an embedding size of 1024. This design allows the context network to model complex dependencies and highlight language-specific characteristics, making it easier to detect inconsistencies.

In conjunction with audio features F_a, which are extracted via Wav2Vec2, we also extract visual features F_v using a TSN backbone. These visual and audio features are concatenated and fed into the TFL detector and refinement module for comprehensive training. This combined approach leverages the robust language adaptability of Wav2Vec2 to enhance generalization, ensuring precise forgery localization by effectively utilizing both visual and audio cues.

4.3 TFL detector

In CLFormer, the TFL detector consists primarily of a multi-level transformer encoder and a feature decoding

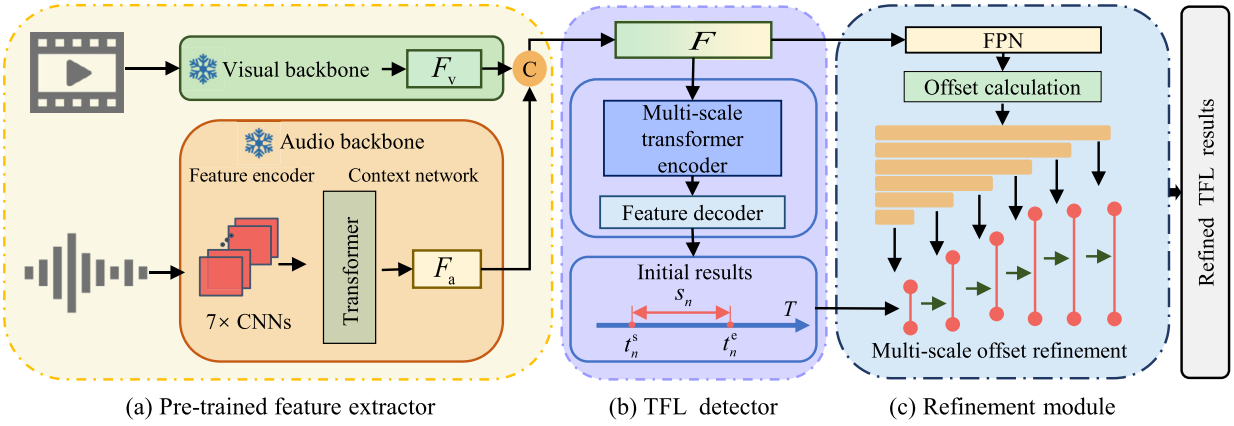

Figure 3 Overview of the cross-lingual transformer framework (CLFormer). The pre-trained feature extractor is used for visual and audio feature extraction, the TFL detector is used for initial forgery detection and localization, and the refinement module is used for fine-tuning localization results. CNNs refer to convolutional neural networks; FPN denotes the feature pyramid network; and TFL stands for temporal forgery localization. F_v: visual features, F_a: audio features, and F: audio-visual features. C represents the concatenated feature dimension. T denotes the temporal segments, t_n^s/t_n^e are the start/end timestamps of the n-th forged segment, and s_n is the confidence score

module. The multi-level transformer encoder enhances the audio-visual features F and encodes them into a feature pyramid. These encoded features are then decoded by the feature decoding module to localize forged segments, resulting in the initial detection output $S = \{(t_n^s, t_n^e, s_n)\}_{n=1}^N$, where t_n^s and t_n^e are the start and end times of the n-th forged segment, s_n is the corresponding confidence score, and N is the total number of detected forged segments.

4.4 Refinement module

Forgery regions often closely resemble authentic regions both visually and auditorily, with ambiguous boundaries that challenge the traditional TFL methods. This complexity is heightened in cross-lingual tasks, where linguistic variations significantly influence the characteristics of forgeries. To address these challenges, we introduce a refinement module in CLFormer, inspired by RefineTAD [25]. This module refines initial results through multi-scale feature processing and precise boundary adjustments, enhancing localization accuracy and generalization, particularly in complex cross-lingual scenarios. The refinement module operates in two steps.

Step 1: Offset calculation We first employ a feature pyramid network (FPN) to encode video features, generating multi-scale representations. For each time point t, pyramidal offsets $\Phi = \{O, C, P\}$ are directly predicted to refine the initial predictions. O represents the offset to the nearest ground truth boundary, C denotes the probability of being fake or real, and P indicates whether point t is close to the boundary of a forged region.

Given the ground truth set $\hat{S} = \{(\hat{t}_n^s, \hat{t}_n^e, \hat{s}_n)\}_{n=1}^N$, where $\hat{t}_n^s$ and $\hat{t}_n^e$ denote the start and end time points of the n-th forged segment, and $\hat{s}_n$ represents the corresponding confidence score. We determine I_s/I_e as the indices of the forged segments that are closest to the time point t by calculating the proximity to the start and end boundaries as

$$I_s = \arg\min_n \left\{\text{abs}(\hat{t}_n^s - t)\right\}_{n=1}^N, \tag{1}$$

$$I_e = \arg\min_n \left\{\text{abs}(\hat{t}_n^e - t)\right\}_{n=1}^N. \tag{2}$$

The offsets are computed via Eq. (3) and Eq. (4):

$$\hat{O}_{l,t,s} = (\hat{t}_{I_s}^s - t)/r_l, \tag{3}$$

$$\hat{O}_{l,t,e} = (\hat{t}_{I_e}^e - t)/r_l, \tag{4}$$

where, $\hat{t}_{I_s}^s$ and $\hat{t}_{I_e}^e$ denote the predicted start and end times of the forged segments that are closest to time t. r_l is the receptive field size, which is calculated as $r_l = r_1 \times 2^{l-1}$, where r_1 is the initial size and l is the pyramid level. We predict C_t, the probability of t being fake or real, and P_t, the boundary-aware probability. The probability P_t is determined by comparing the predicted offset $\hat{O}_t$ with the receptive field r_l. Specifically, if $|\hat{O}_t| \leq r_l$, t is considered likely close to the boundary; otherwise, it is less likely to be close to the boundary.

Step 2: Multi-scale offset refinement We refine the initial detection results $S = \{(t_n^s, t_n^e, s_n)\}_{n=1}^N$ using a multi-scale approach. The offsets Φ from Step 1 are utilized for further enhancement. For each forged segment, the offsets are calculated based on their start time t^s and end time t^e using

their positions in different layers. The calculations are as

$$\{o_s, c_s, p_s\} = \omega \times \Phi_{l,\alpha_s,t^s} + (1-\omega) \times \Phi_{l,\beta_s,t^s}, \tag{5}$$

$$\{o_e, c_e, p_e\} = \omega \times \Phi_{l,\alpha_e,t^e} + (1-\omega) \times \Phi_{l,\beta_e,t^e}, \tag{6}$$

where o_s, c_s, p_s and o_e, c_e, p_e represent the offset, confidence, and boundary-aware probability for the start and end times of the forged segment. Φ_{l,α_s,t^s} and Φ_{l,β_s,t^s} are the left and right pyramidal offsets for the start time t^s, calculated at level l, where $\alpha_s = \lfloor t^s/r_l \rfloor$ and $\beta_s = \lceil t^s/r_l \rceil$; similarly, Φ_{l,α_e,t^e} and Φ_{l,β_e,t^e} are the pyramidal offsets for the end time t^e, with $\alpha_e = \lfloor t^e/r_l \rfloor$ and $\beta_e = \lceil t^e/r_l \rceil$. $\omega = \beta_s - t^s/r_l$ means the left weight, and $1-\omega$ represents the right. We refine the start and end times as

$$t^{s'} = t^s + o_s \times r_l \times (1-s), \tag{7}$$

$$t^{e'} = t^e + o_e \times r_l \times (1-s). \tag{8}$$

Next, we update the confidence score s based on the refined probabilities:

$$s' = s \times (1+c_s) \times p_s \times (1+c_e) \times p_e. \tag{9}$$

These steps refine the initial detection results, leading to the final output $S' = \{(t_n^{s'}, t_n^{e'}, s_n')\}_{n=1}^N$. This refined output enhances localization accuracy and generalization in CLFormer, ensuring robust performance across various linguistic contexts.

4.5 Training and inference

CLFormer is trained in two stages. In the first stage, similar to UMMAFormer [6], the following loss is employed for initial detection:

$$L_{\text{TFL}} = L_{\text{cls}} + \lambda_{\text{reg}} L_{\text{reg}} + \lambda_{\text{rec}} L_{\text{rec}} + \lambda_{\text{scls}} L_{\text{scls}}, \tag{10}$$

where L_{cls}, L_{reg}, L_{rec}, and L_{scls} represent the focal, regression, reconstruction, and sample-level focal losses, respectively; the hyperparameters λ_{reg}, λ_{rec}, and λ_{scls} balance their contributions. Soft non-maximum suppression is employed to remove redundant predictions and obtain initial results.

In the second stage, the refinement module is trained. Its loss function comprises L1 loss L_{ref}, L_{cls}, and smooth-L1 loss L_{prob}.

$$L_{\text{RM}} = \frac{1}{N_{\text{pos}}} \sum \mathbf{1}_{\{\hat{P}=1\}} \left(L_{\text{ref}} + L_{\text{cls}}\right) + \frac{1}{N} \sum L_{\text{prob}}, \tag{11}$$

where N_{pos} and N are the numbers of positive and total samples, respectively. $\mathbf{1}_{\{\hat{P}=1\}}$ indicates that only time points around the ground-truth boundary are focused on during the training process. During inference, a multi-scale strategy is applied to refine the results and increase the precision and confidence scores.

5 Experiment

5.1 Experiment setting

Datasets We evaluate CLFormer on the LAV-DF [3], AV-Deepfake1M [7], and CHAV-DF datasets. LAV-DF includes 136,304 videos, including 36,431 authentic and 99,873 fake videos across 153 identities. AV-Deepfake1M contains 1,146,760 videos, with 286,721 authentic videos and 86,039 partially forged videos. Owing to the size limitation of the dataset, we re-split the data by extracting 200,000 samples from the training set for training and using its validation set as the test set.

Evaluation metrics We use AP and average recall (AR) as evaluation metrics, with tIoU thresholds of 0.50, 0.75, and 0.95 for the AP and proposals of 10, 20, 50, and 100 for the AR. We also compute the mean values of AP and AR to provide a more comprehensive result.

Implementation details We train at 25 fps with parameters consistent with those of UMMAFormer [6]. To ensure fairness in the experiments, we down-sampled the CHAV-DF at a resolution of 640×360, with a frame rate of 25 fps. Additionally, the audio data is sampled at a frequency of 16 kHz. For feature extraction, TSN pre-trained on ActivityNet [26] is employed for 4096-dimensional video features, including both RGB and optical flow features, and Wav2Vec2 is employed for 1024-dimensional audio features.

5.2 Comparison with state-of-the-art methods

We evaluate the performance of state-of-the-art methods in intra-domain settings, with the experimental results shown in Tables 1 and 2. As displayed in Table 1, on LAV-DF, CLFormer achieves an AP of 99.33% at tIoU of 0.50 and an AP of 70.07% at tIoU of 0.95. These results outperform the second-best method, VIGO [13], by 2.71% at tIoU of 0.50 and by impressive 5.03% at tIoU of 0.95, and a 3.51% improvement in the average AR. As displayed in Table 2, on AV-Deepfake1M, CLFormer demonstrates a significant improvement over the second-best method, UMMAFormer [6], with a 14.42% increase in AP at tIoU of 0.95 and a 6.71% increase in the average AP. On CHAV-DF, CLFormer achieves an impressive 8.58% improvement in the AP at tIoU of 0.95 compared with UMMAFormer, further highlighting its superior precision and effectiveness in the single-language TFL. Table 1 and Table 2 reveal that CLFormer performs better on CHAV-DF than on LAV-DF and AV-Deepfake1M. This is mainly because of the influence of the Chinese recitation speech rate, which results in longer forged segments that lead to better detection performance by the model.

To validate the generalization ability of CLFormer, we evaluate the performance of state-of-the-art methods on

Table 1 Performance comparison on the LAV-DF [3] dataset. AP and AR indicate the average precision and average recall, respectively. Avg. indicates the mean value computed across the different thresholds or proposal counts. The best results are highlighted in bold

Train & test		LAV-DF								
Method	Audio feature	AP (%)				AR (%)				
		0.50	0.75	0.95	Avg.	100	50	20	10	Avg.
BA-TFD [3]	Audio	76.90	38.50	0.25	38.55	66.90	64.08	60.77	58.42	62.54
BA-TFD+ [4]	Audio	96.30	84.96	4.44	61.90	81.62	80.48	79.40	78.75	80.06
ActionFormer [27]	BYOL-A [28]	97.57	93.81	40.34	77.24	92.88	92.87	92.70	92.25	92.68
AV-TFL [9]	Audio	94.89	74.87	1.94	57.23	76.12	74.24	72.17	71.24	73.44
Tridet [29]	BYOL-A	95.60	87.48	25.65	69.58	89.86	89.82	89.42	88.51	89.40
UMMAFormer [6]	BYOL-A	98.83	95.54	37.61	77.33	92.48	92.47	92.42	92.10	92.37
DiMoDif [10]	Ma et al. [30]	95.50	87.90	20.60	68.00	94.20	93.70	92.70	91.40	93.00
MMMS-BA [11]	Audio	97.56	95.25	39.02	77.28	94.05	93.45	95.93	89.42	93.21
VIGO [13]	Wav2Vec2 [8]	96.62	94.05	65.04	85.24	93.34	93.34	93.34	93.34	93.34
CLFormer	Wav2Vec2	**99.33**	**97.43**	**70.07**	**88.94**	**96.91**	**96.91**	**96.88**	**96.69**	**96.85**

both cross-language domain and cross-same-language domain. Specifically, we train the model separately on LAV-DF, CHAV-DF, and AV-Deepfake1M, and test it on the remaining two unseen datasets. We observe a significant decline in performance when the language setting and dataset change. The results in Table 3 and Table 4, tested on datasets from different languages, demonstrate that existing methods experience a more significant performance drop in cross-language domain, compared with cross-same-language domain.

In the comparison of cross-same-language domain, Table 3 shows the performance of CLFormer when trained on LAV-DF and tested on AV-Deepfake1M, which achieves an average AP of 34.66% and an average AR of 53.06%. CLFormer reaches an AP of 59.88% at tIoU of 0.50, showing an improvement of 48.90% over the second-best method, Tridet [29]. In Table 4, we observe that CLFormer, when trained on AV-Deepfake1M and tested on LAV-DF, achieves an AP of 83.63% at tIoU of 0.50, surpassing the second-best method, Tridet, by 26.26%.

In the comparison of cross-language domain, Table 3 demonstrates that CLFormer, trained on LAV-DF and tested on CHAV-DF, achieves an average AP of 13.06% and an average AR of 56.72%, outperforming UMMAFormer by 9.90% and 25.42%, respectively. In Table 4, when trained on AV-Deepfake1M and tested on CHAV-DF, CLFormer achieves an average AP of 8.93% and an average AR of 41.14%, surpassing Tridet by 0.43% and 3.10%, respectively. Table 5 shows that CLFormer outperforms all other methods across all the evaluation metrics, both on LAV-DF and AV-Deepfake1M. Overall, CLFormer exhibits exceptional generalization ability, achieving the best performance in both the cross-language and cross-same-language domains.

To further analyze the performance of different methods in both the cross-same-language domain and the cross-language domain, we quantify the performance degradation by calculating relative performance drops (Avg. AP drop and Avg. AR drop), where lower values indicate better performance. The models are trained on LAV-DF and CHAV-DF, and tested on AV-Deepfake1M. The results are presented in Table 6. The results show that performance drops are more pronounced in cross-language settings, highlighting the challenges of cross-language tasks. Specifically, when CLFormer is trained on LAV-DF and tested on AV-Deepfake1M, its Avg.AP drop is 31.90% lower and its Avg.AR drop is 17.72% lower than Tridet. In the cross-language setting (CHAV-DF trained, AV-Deepfake1M tested), CLFormer exhibits a 10.61% lower Avg.AP drop and a 19.09% lower Avg.AR drop than Tridet does. These results demonstrate the excellent performance and generalization capabilities of CLFormer in cross-same-language domain and cross-language domain, confirming its effectiveness as a cross-language forgery detection solution.

5.3 Ablation studies

To evaluate the impact of different audio feature extraction methods on TFL, we specifically compare BYOL-A [28], linear frequency cepstral coefficients (LFCC), and Wav2Vec2 [8] features. As illustrated in Fig. 4, incorporating Wav2Vec2 features result in a substantial improvement in the accuracy of the localization, particularly when the training and testing domain are aligned. This improvement is pronounced in the AP at tIoU of 0.95, where Wav2Vec2 features enable more precise localization. Moreover, these features significantly increase the recall rate and reduce the number of false positive detections.

In cross-language domain and cross-same-language domain experiments, Wav2Vec2 features consistently outperform the previously used BYOL-A features, as reflected in substantial improvements in the AP at tIoU of 0.50 and AR metrics. These results highlight the generalization capability of Wav2Vec2, demonstrating its effectiveness in capturing diverse audio characteristics. This performance

Table 2 Performance comparison on the AV-Deepfake1M [7] and Chinese audio-visual deepfake (CHAV-DF) datasets when trained and tested on the same dataset

Train & test		AV-Deepfake1M									CHAV-DF								
Method	Audio feature	AP (%)				AR (%)					AP (%)				AR (%)				
		0.50	0.75	0.95	Avg.	100	50	20	10	Avg.	0.50	0.75	0.95	Avg.	100	50	20	10	Avg.
ActionFormer	BYOL-A	95.45	89.36	27.48	70.76	91.10	91.02	90.39	88.99	90.38	98.39	96.67	79.02	91.36	97.24	97.24	97.23	97.06	97.19
Tridet	BYOL-A	96.59	90.45	25.67	70.90	90.62	90.56	90.08	89.12	90.10	98.20	96.48	72.06	88.91	96.75	96.75	96.71	96.56	96.69
UMMAFormer	BYOL-A	96.53	91.54	32.41	73.49	91.76	91.70	91.29	90.43	91.30	**98.68**	**97.00**	79.90	91.86	97.31	97.31	97.31	97.16	97.27
CLFormer	Wav2Vec2	**98.67**	**95.10**	**46.83**	**80.20**	**93.98**	**93.97**	**93.84**	**93.36**	**93.79**	98.58	96.87	**88.48**	**94.64**	**98.21**	**98.21**	**98.07**	**97.92**	**98.10**

Table 3 Performance of the models trained on the LAV-DF dataset and tested on the CHAV-DF and AV-Deepfake1M datasets

Train		LAV-DF																	
Test		CHAV-DF									AV-Deepfake1M								
Method	Audio feature	AP (%)				AR (%)					AP (%)				AR (%)				
		0.50	0.75	0.95	Avg.	100	50	20	10	Avg.	0.50	0.75	0.95	Avg.	100	50	20	10	Avg.
ActionFormer	BYOL-A	4.56	0.29	0.00	1.62	27.76	27.63	27.57	24.48	26.86	4.44	1.58	0.01	2.01	24.90	23.53	19.84	15.79	21.02
Tridet	BYOL-A	2.58	0.11	0.00	0.90	20.18	20.15	19.74	16.24	19.08	10.98	3.76	0.02	4.92	37.48	36.52	32.26	26.28	33.14
UMMAFormer	BYOL-A	7.31	2.06	**0.11**	3.16	33.73	33.73	33.28	24.46	31.30	8.28	3.27	0.02	3.86	30.02	29.05	25.81	20.27	26.29
CLFormer	Wav2Vec2	**24.11**	**14.97**	0.09	**13.06**	**58.19**	**58.11**	**57.89**	**52.67**	**56.72**	**59.88**	**44.07**	**0.03**	**34.66**	**54.55**	**54.25**	**52.93**	**50.50**	**53.06**

Table 4 Performance of the models trained on the AV-Deepfake1M dataset and tested on the LAV-DF and CHAV-DF datasets

Train		AV-Deepfake1M																	
Test		LAV-DF									CHAV-DF								
Method	Audio feature	AP (%)				AR (%)					AP (%)				AR (%)				
		0.50	0.75	0.95	Avg.	100	50	20	10	Avg.	0.50	0.75	0.95	Avg.	100	50	20	10	Avg.
ActionFormer	BYOL-A	51.13	33.90	0.19	28.41	61.36	60.78	57.05	52.76	57.99	1.54	0.06	0.00	0.53	15.87	15.86	15.19	11.51	14.61
Tridet	BYOL-A	57.37	50.70	1.44	36.50	73.38	73.28	71.86	68.71	71.81	**14.55**	9.93	1.01	8.50	40.79	40.79	38.83	31.75	38.04
UMMAFormer	BYOL-A	56.91	49.72	2.21	36.28	74.55	74.23	71.46	67.29	71.88	12.69	9.02	1.69	7.80	43.08	42.40	33.15	26.22	36.21
CLFormer	Wav2Vec2	**83.63**	**76.96**	**4.38**	**54.99**	**81.15**	**81.10**	**80.47**	**78.82**	**80.39**	13.39	**10.36**	**3.05**	**8.93**	**45.72**	**45.48**	**39.84**	**33.50**	**41.14**

Table 5 Performance of the models trained on the CHAV-DF dataset and tested on the LAV-DF and AV-Deepfake1M datasets

Train		CHAV-DF																	
Test		LAV-DF									AV-Deepfake1M								
Method	Audio feature	AP (%)				AR (%)					AP (%)				AR (%)				
		0.50	0.75	0.95	Avg.	100	50	20	10	Avg.	0.50	0.75	0.95	Avg.	100	50	20	10	Avg.
ActionFormer	BYOL-A	3.11	1.33	0.03	1.49	49.97	48.06	37.26	27.46	40.69	1.17	0.24	0.00	0.47	29.18	25.80	15.09	9.83	19.98
Tridet	BYOL-A	10.09	5.40	0.21	5.23	36.84	36.63	34.36	30.06	34.47	1.64	0.34	0.01	0.66	22.83	21.80	17.18	11.54	18.34
UMMAFormer	BYOL-A	3.48	1.49	0.08	1.68	38.50	38.42	34.08	24.61	33.90	0.85	0.18	0.00	0.34	17.40	17.13	12.41	7.37	13.58
CLFormer	Wav2Vec2	**57.68**	**49.80**	**3.64**	**37.04**	**69.82**	**69.81**	**65.46**	**60.88**	**66.49**	**18.13**	**13.54**	**0.56**	**10.74**	**42.13**	**42.03**	**35.16**	**30.04**	**37.34**

Table 6 Cross-domain and cross-language performance drops of the models trained on the LAV-DF and CHAV-DF datasets and tested on the AV-Deepfake1M dataset. ↓ indicates the lower value is better

Train	LAV-DF		CHAV-DF	
Method	Avg.AP drop ↓ (%)	Avg.AR drop ↓ (%)	Avg.AP drop ↓ (%)	Avg.AR drop ↓ (%)
ActionFormer	97.40	77.32	99.49	79.44
Tridet	92.93	62.93	99.26	81.03
UMMAFormer	95.01	71.54	99.63	86.04
CLFormer	**61.03**	**45.21**	**88.65**	**61.94**

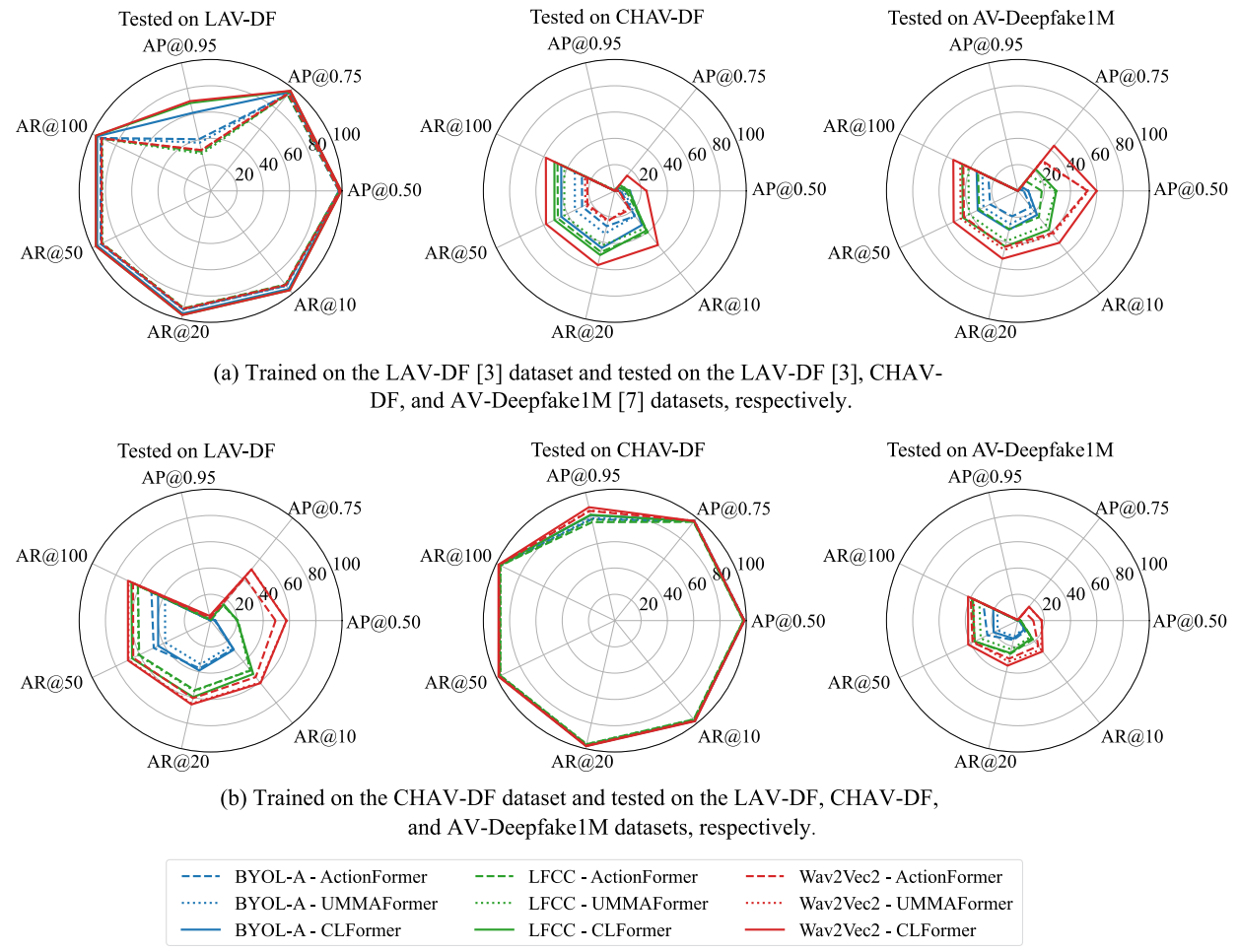

Figure 4 Evaluation of the effect of audio feature extraction methods (BYOL-A [28], linear frequency cepstral coefficients (LFCC), and Wav2Vec2 [8]) on the TFL performance across multiple models. The comparison includes three models: ActionFormer [27], UMMAFormer [6], and CLFormer. We use average precision (AP) and average recall (AR) as evaluation metrics. AP@0.50, AP@0.75, and AP@0.95 represent the average precision at temporal intersection over union (tIoU) thresholds of 0.50, 0.75, and 0.95, respectively. AR@10, AR@20, AR@50, and AR@100 indicate the average recall when using a maximum of 10, 20, 50, and 100 proposals, respectively. Different colors represent different audio features, while different line styles correspond to different model architectures

establishes Wav2Vec2 as a robust solution for the multi-language and cross-dataset TFL.

To assess the impact of the introduced refinement module, we compare performance metrics before and after its integration across various datasets and languages, as shown in Fig. 5. The results reveal a significant enhancement in precision because of the refinement module. Notably, the module yields a substantial increase in the AP at tIoU of 0.95, reflecting a marked improvement in the localization accuracy. Additionally, it consistently increases

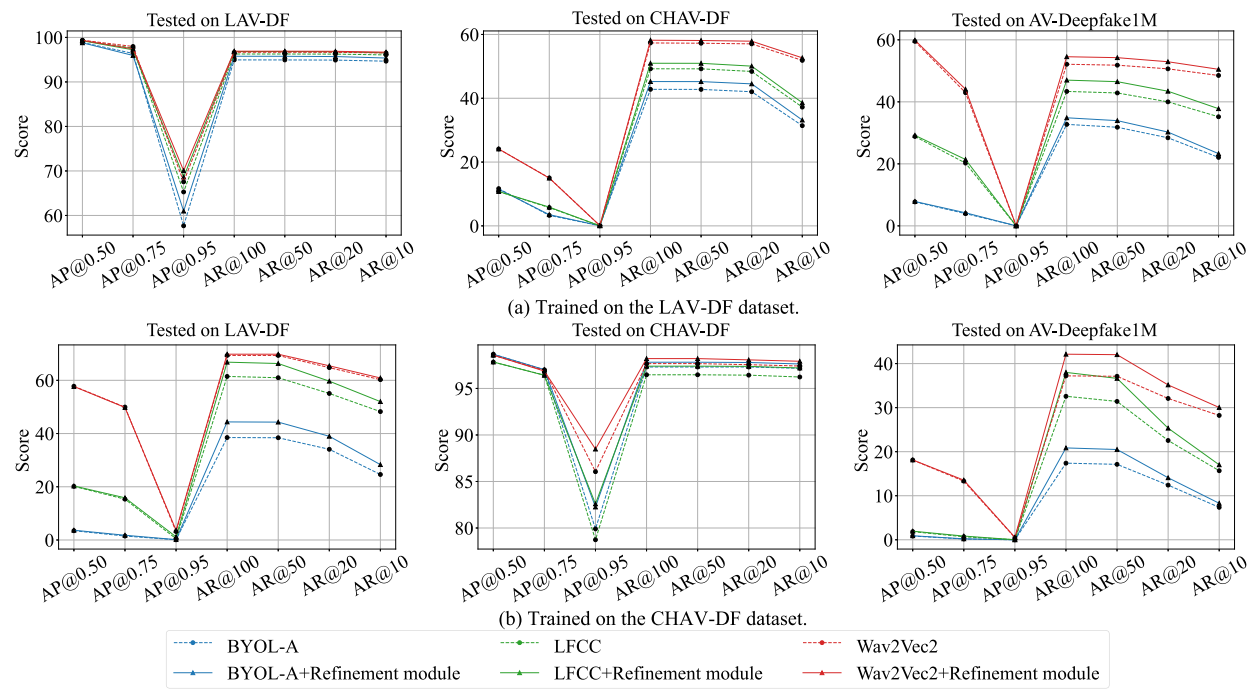

Figure 5 Effect of the refinement module on the CLFormer performance in intra-domain and cross-domain settings under different audio feature configurations

the recall rate, effectively reducing the incidence of false negatives. These findings demonstrate that the refinement module not only enhances the accuracy of TFL but also significantly improves generalization and robustness across different datasets and languages.

5.4 Visualization results

To qualitatively assess the efficacy of CLFormer, we present the results of models trained on CHAV-DF and evaluated them on various datasets, as illustrated in Fig. 6. The visualization results reveal a marked degradation in localization performance for existing methods in cross-lingual scenarios. While the baseline model presents significant challenges in accurate localization, CLFormer maintains high precision. These observations underscore the superior effectiveness of CLFormer in the cross-lingual TFL.

6 Conclusion and future work

In this paper, we introduce CHAV-DF, which sets a new benchmark for TFL in non-English languages. We also present the CLFormer, a cross-lingual transformer framework integrating Wav2Vec2 and a refinement module. This novel approach significantly enhances both TFL accuracy and cross-lingual generalization. However, challenges remain in achieving high-precision temporal localization and the effective handling of increasingly sophisticated and complex forgery scenarios.

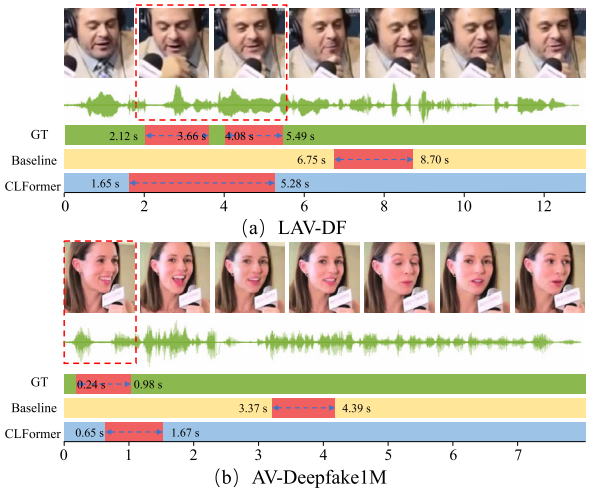

Figure 6 Visualization of cross-lingual performance, with red indicating fake segments and green representing true segments. The images are sourced from the LAV-DF and AV-Deepfake1M datasets. GT refers to ground truth

Future work will focus on expanding the dataset to include a broader range of languages from different linguistic families, thereby further enhancing the ability of the model to generalize across languages. Additionally, we aim to refine the audio-visual fusion mechanisms to enhance the ac-

curacy and reliability of temporal localization, particularly when identifying subtle and varied deep forgery patterns.

Abbreviations
AP, average precision; AR, average recall; CHAV-DF, Chinese audio-visual deepfake dataset; CLFormer, cross-lingual transformer framework; CNNs, convolutional neural networks; FPN, feature pyramid network; LFCC, linear frequency cepstral coefficients; RNNs, recurrent neural networks; TFL, temporal forgery localization; tIoU, temporal intersection over union; TTS, text-to-speech.

Author contributions
All the authors contributed to the study conception and design. Conceptualization of this study, methodology: HC; material preparation, methodology: HL; methodology: JC; investigation, validation: LY. All the authors read and approved the final manuscript.

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Data availability
The datasets used in this study are publicly available at: AV-Deepfake1M: https://github.com/ControlNet/AV-Deepfake1M; LAV-DF: https://github.com/ControlNet/LAV-DF.

Declarations

Competing interests
The authors declare that they have no competing interests.

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